

The Climate System

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- Issues with homework?
- For water balance problem, assume steady-state (no change in storage)
- Typo on page 6, p_D should be p_L for lake precip.
- Textbook edition 3 available in bookstore and on RedShelf, but Heather found a free PDF of 2nd edition online!

- capillary rise problem, f_{lake} is relative (dimensionless)

Corrections Corner

- **Primary** dimensions are Mass (M), Length (L), Time (T).
- Force, pressure, and energy, often expressed in equations as F, P, and E, respectively, are derived from primary dimensions of M, L, T

State Variables

A state variable is a measurable physical quantity that defines the state or condition (at a specific location and time) of Earth's system.

We are interested in these because they tell us something about the system at any given point in time and space, particularly how they change over space or time. We are also interested in predicting what these states will be in the future...

<https://www.nhc.noaa.gov/>

Pressure-Temperature Relations

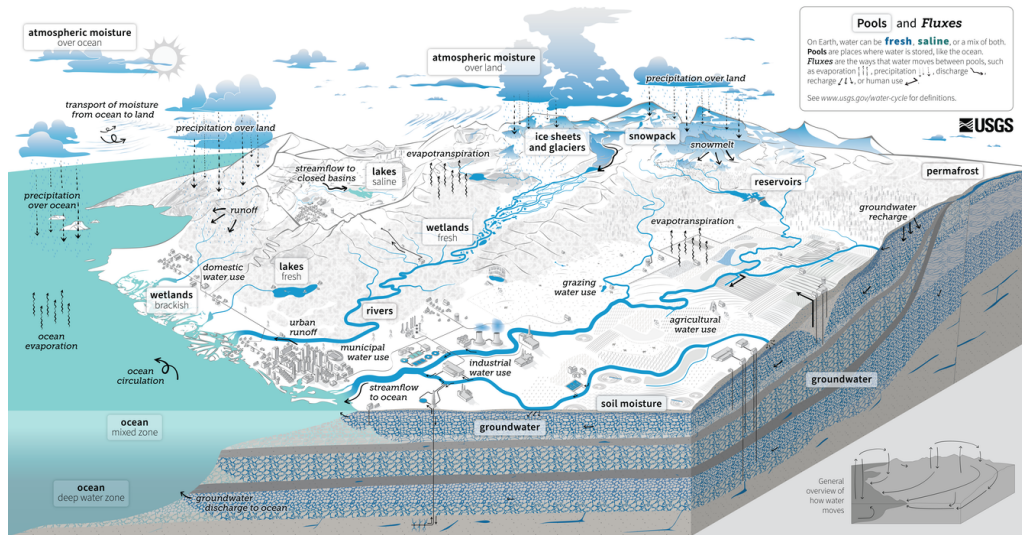
The variations with pressure, temperature, and density with altitude in the atmosphere can be explain by the Ideal Gas Law:

$$PV = nRT$$

where P is atmospheric pressure in Pa, V is volume of air parcel in m^3 , n is the amount of substance in moles, R is the gas constant in $Pa \cdot m^3 \cdot mol^{-1} \cdot K^{-1}$, T and is air temperature in K.

These are the major state variables of the atmosphere (not including water vapor, which we will discuss in the next lecture).

Water Cycle



What is the difference between climate and weather?

What is the difference between climate and weather?

Weather (meteorology): state (e.g., temperature, humidity, wind speed, etc.) of the atmosphere at a given time/place.

Climate (climatology): average of weather in terms of mean state and its variability over defined time period/region.

Learning Objectives

- 1 Understand the physics of radiation and apply it to the Sun/Earth.
- 2 Understand Earth's energy balance.
- 3 Understand how the uneven distribution of radiation drives general circulation.
- 4 How radiation and general circulation (among other factors) strongly control climate.

Physics of Radiant Energy

All matter above absolute zero (0 K) radiates energy in the form of electromagnetic waves that travel at the speed of light. The rate at which this energy is emitted is given by Stefan Boltzmann Law:

$$Q_r = \varepsilon \cdot \sigma \cdot T^4$$

where Q_r is rate of energy emission per unit surface area per unit time, T is absolute temperature of surface, σ is the Stefan Boltzmann constant ($5.67 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$), and ε is emissivity of the surface (dimensionless). The value of ε ranges from 0 to 1, with 1 being referred to as a **blackbody**.

From this, what state variable controls energy emission?

Emissivity

Ranges from 0 to 1, with 0 acting as a perfect reflector and 1 as a perfect emitter.

TABLE 3.1. TYPICAL BROADBAND EMISSIVITIES FOR VARYING LAND SURFACE TYPES (FROM ARYA, 2001)

SURFACE TYPE	EMISSIVITY
Water	0.92-0.97
Snow	0.82-0.99
Ice	0.92-0.97
Bare soil	0.84-0.97
Grass (long, 1m)	0.90
Grass (short, 0.02m)	0.95
Agricultural crops	0.90-0.99
Forests	0.97-0.99

Albedo

Ranges from 0 to 1, with 0 absorbing all incoming radiation and 1 reflecting all.

TABLE 3.2. TYPICAL BROADBAND ALBEDO FOR VARYING LAND SURFACE TYPES (FROM ARYA, 2001)

SURFACE TYPE	ALBEDO
Water (small zenith angle)	0.03-0.10
Water (large zenith angle)	0.10-1.0
Snow (old)	0.40-0.70
Snow (fresh)	0.45-0.95
Ice	0.20-0.45
Bare soil	0.05-0.40
Grass (long, 1m)	0.16
Grass (short, 0.02m)	0.26
Agricultural crops	0.10-0.25
Forests	0.05-0.20

Relationship of Radiative Properties

For opaque (non-transparent) objects in thermodynamic equilibrium, emissivity is assumed equal to absorptivity and inversely related to albedo (reflectivity):

$$\varepsilon = 1 - \alpha$$

where ε is emissivity and α is albedo.

Does this comport with the ε and α values on the previous slides?

The Sun

The Sun radiates energy as a blackbody with a temperature of 5773 K.

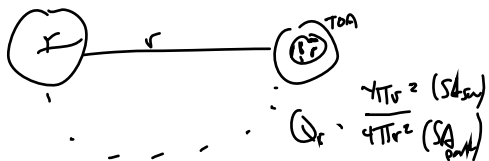
What is the total energy emitted at the sun's surface, and what is the rate of energy delivered to the top of earth's atmosphere (TOA)?

$$Q_r = \epsilon \sigma \cdot T^4 \quad \sigma = 5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}$$

$$R_{\text{sun}} = 6.957 \times 10^8 \text{ m}$$

$$r_{\text{sun-earth}} = 1.5040 \times 10^{11} \text{ m}$$

$$I_0 = 1,346.1 \frac{\text{W}}{\text{m}^2}$$



$$1367$$

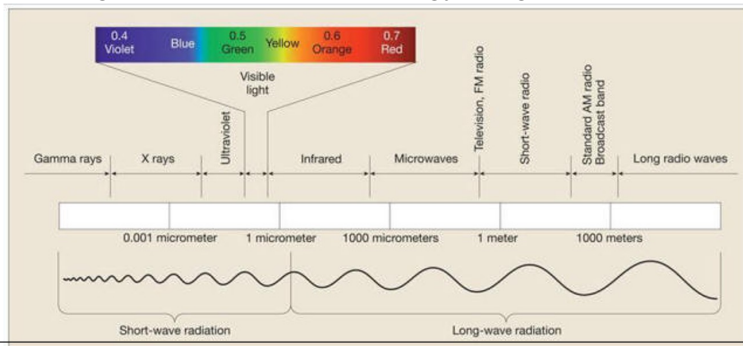
Physics of Radiant Energy

Wavelength and frequency of electromagnetic radiation are inversely related:

$$\lambda \cdot f = c$$

where λ is wavelength and f is frequency, and c is speed of light (2.998×10^8 m/s).

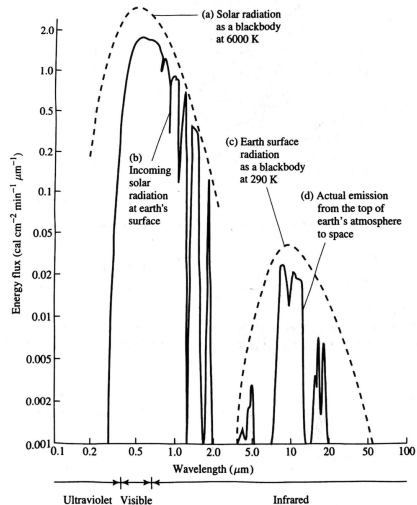
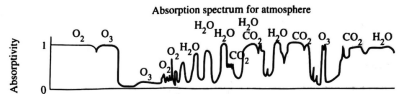
λ (measured in μm) varies over 21 orders of magnitude, but mostly the infrared, visible, and ultraviolet wavelengths matter for Earth energy budget.



Solar Radiation

Gases in the atmosphere absorb much of the lower wavelength (ultraviolet) radiation, which is damaging to most forms of life.

What arrives at the Earth's surface is called **solar** or **shortwave radiation**



Composition of Atmosphere

TABLE 2.1. PRIMARY CONSTITUENTS OF THE
STANDARD ATMOSPHERE IN THE TROPOSPHERE

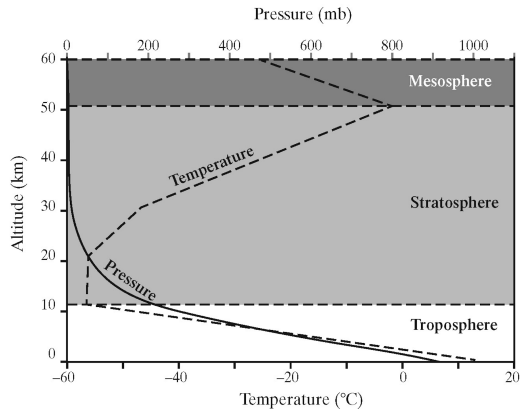
GAS	% BY VOLUME
Nitrogen (N ₂)	78.084
Oxygen (O ₂)*	20.946
Argon (A)	0.934
Carbon Dioxide (CO ₂)*	0.040
Neon (Ne)	0.00182
Helium (He)	0.000524
Methane (CH ₄)*	0.00016
Krypton (Kr)	0.0014
Hydrogen (H ₂)	0.00005
Nitrous Oxide (N ₂ O)	0.000035
Ozone (O ₃)*	0 - 0.000007
Water (H ₂ O)*	0 - 4

*radiatively active gases

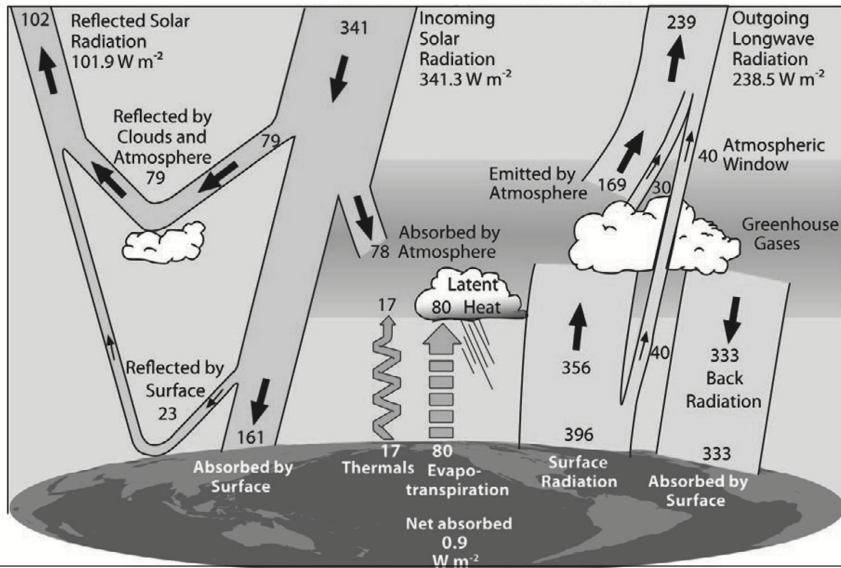
Structure of Atmosphere

Pressure, density, and temperature decrease with increasing altitude, with one exception: temperature in stratosphere. Here, temperature increases from tropopause to stratopause due to the presence of ozone (O_3)

Troposphere and stratosphere are only spheres that directly impact climate, weather, and hydrologic activity.



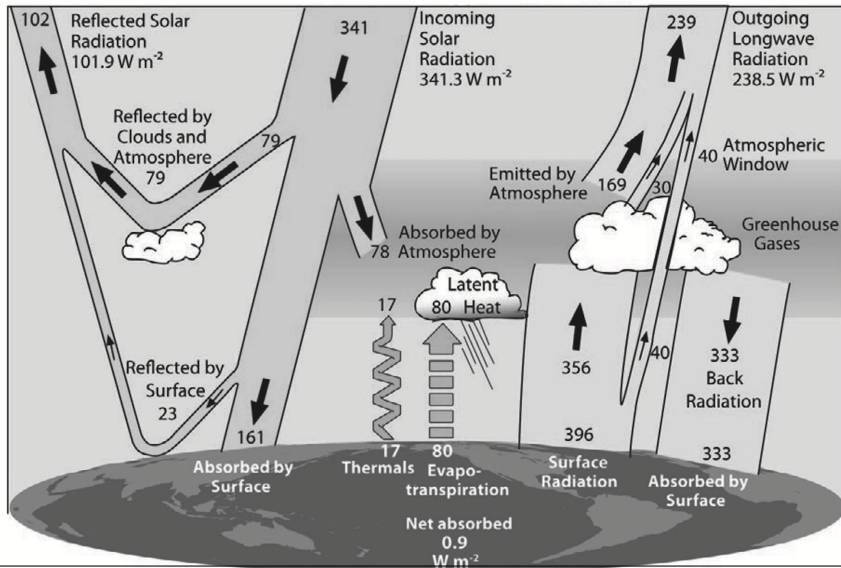
Earth Energy Budget



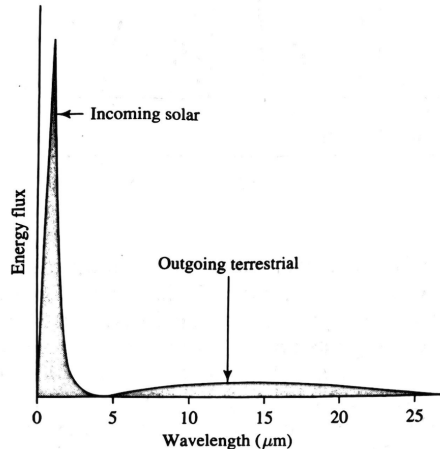
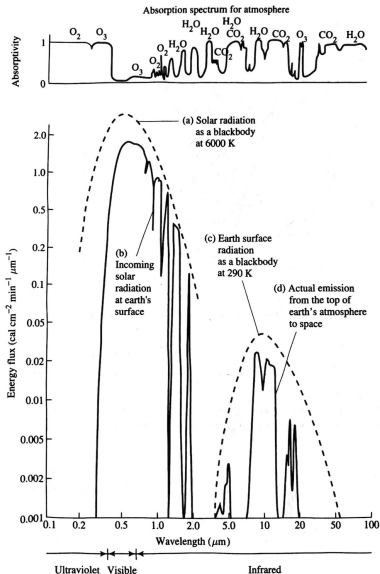
Latent and Sensible Heat Transfer

- 1 **Latent** heat transfer is energy stored or released due to a phase change, in this case, mainly evaporation of water from oceans.
- 2 **Sensible** heat transfer is from conduction/convection from earth's surface to warm the overlying air.

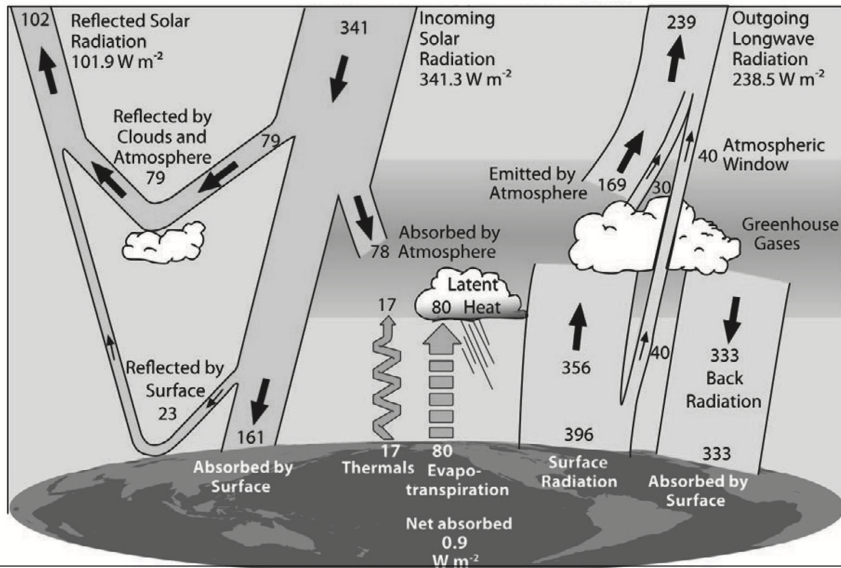
Earth Energy Budget



Radiation, revisited



Earth Energy Budget



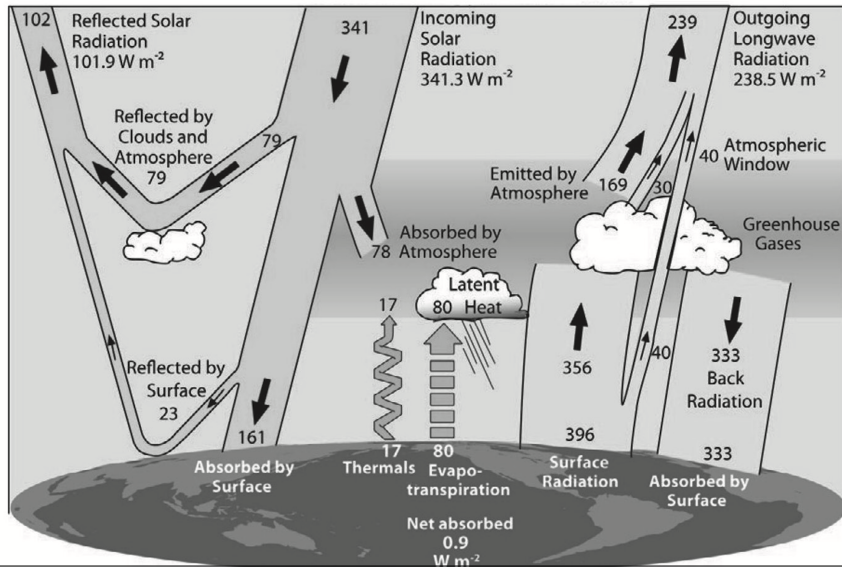
Greenhouse Effect

Absorption and downward re-radiation of longwave radiation is the **greenhouse effect**, which controls the temperature of Earth's surface and atmosphere.

Water vapor is most important contributor followed by clouds and *non-condensing* greenhouse gases: carbon dioxide (CO_2), ozone (O_3), nitrous oxide (N_2O), and methane (NH_3).

Fun fact: while greenhouse gases only *directly* contribute to 25% of greenhouse effect, the warming induced by higher concentrations of them creates a positive feedback, where warmer air holds more water vapor and clouds.

Conservation of Energy

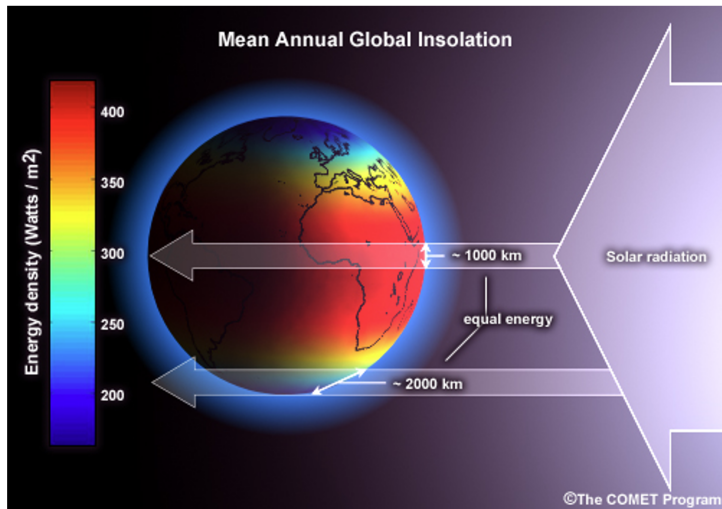


Factors affecting Insolation

341 W m^{-2} is an average rate of solar (SW) radiation, what factors might control variability in that value?

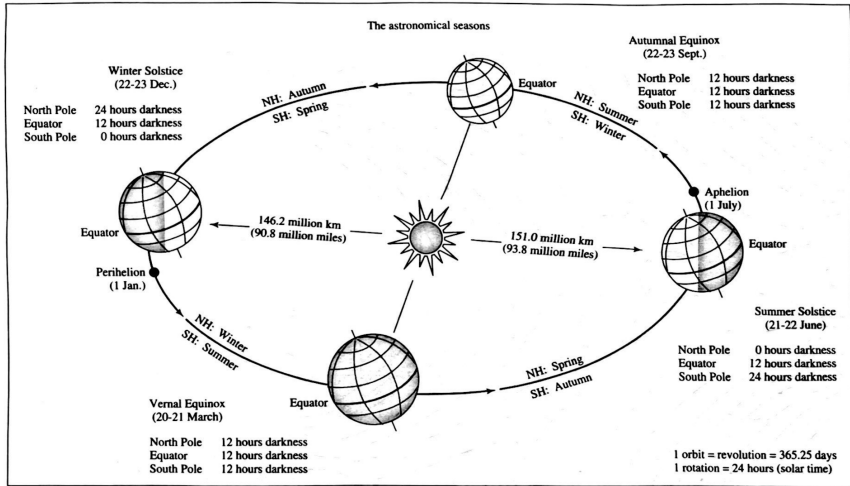
- earth is a sphere
 - earth is spinning
 - earth is orbiting (and orbit is elliptical)
 - earth is tilted
 - altitude
 - albedo / cloud cover
 - aspect
 - slope
- } geometric controls

Latitudinal Variation of TOA SR



Lower intensity at higher latitudes (same energy spread over larger area).

Seasonal Variation of TOA SR



Tilt of 23.5° relative to orbital plane combined with revolution around sun creates seasons.

Estimating SW Radiation at TOA

The solar radiative flux (R_{s0}) at the TOA is as follows (assuming daytime, at nighttime R_{s0} is 0)

$$R_{s0} = I_0 \frac{\cos(\theta)}{r^2}$$

$$\cos(\theta) = \sin(\delta) \cdot \sin(\lambda) + \cos(\delta) \cdot \cos(\lambda) \cdot \cos(\tau)$$

$$\delta = \frac{23.45\pi}{180} \cdot \cos\left[\frac{2\pi}{365} \cdot (172 - DOY)\right]$$

$$\tau = 2\pi \cdot \frac{T_h - 12}{124}$$

$$r = 1 + 0.017 \cdot \cos\left[\frac{2\pi}{365} \cdot (186 - DOY)\right]$$

I_0 = solar constant (1367 W/m²)

θ = solar zenith angle (radians) δ = declination angle (radians)

DOY = day of year

λ = latitude (radians)

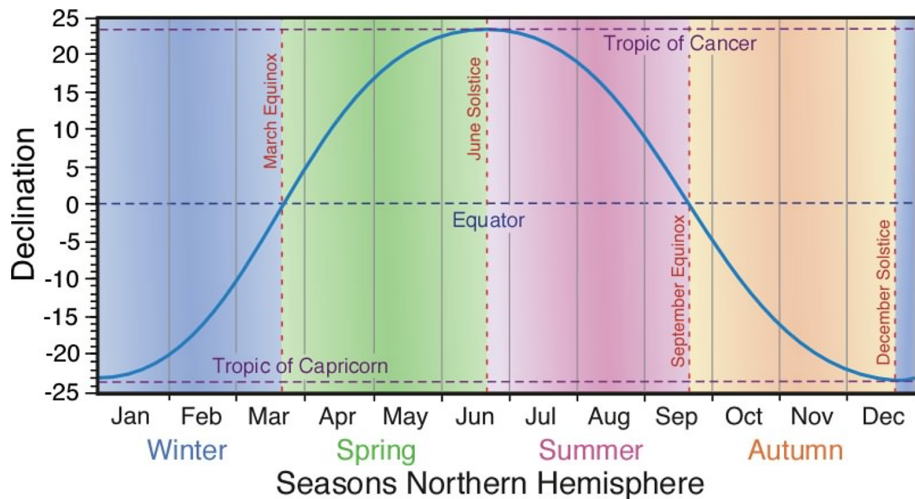
τ = hour angle (radians)

T_h = hour of day (0..23)

r = ratio of actual to mean Earth-Sun distance (-)

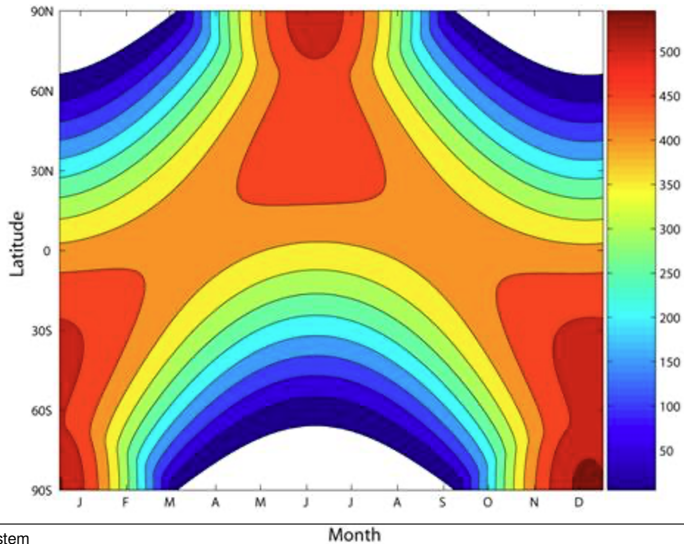
Declination

Declination: latitude at which the sun is directly overhead at noon.

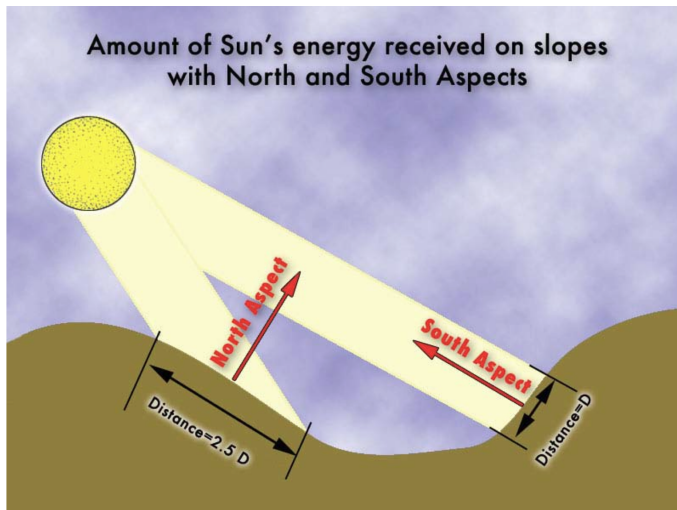


TOA SR Variation

Daily-averaged TOA SW Flux (W m^{-2})



Topographic Controls



Topographic Controls

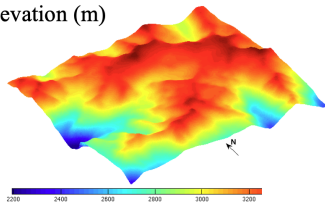
More densely vegetated
(north facing)

Less densely vegetated
(south facing)

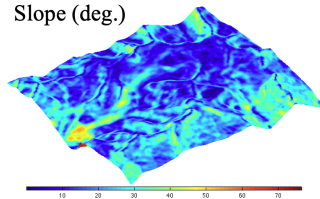


Topographic Controls

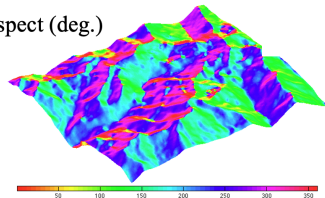
Elevation (m)



Slope (deg.)



Aspect (deg.)



Diffuse Radiation

SW radiation can be further broken down into: direct and diffuse, where direct is from roughly parallel beams to the surface and diffuse has been scattered in the atmosphere and reaches the surface from multiple directions.



Measuring Radiation

SW radiation can be measured in situ using pyranometers, as well as quantified using satellites data.



<https://nsrdb.nrel.gov/data-viewer>